Chapter 8 exercises

Vectors are columns unless stated otherwise. For each coordinate matrix, state the domain basis, codomain basis, and coordinate order.

CH08-E1

Decide linearity for each map. Prove it or give a counterexample.

1. $T_1(x, y) = (2x - y, x + 3y)$.
2. $T_2(x, y) = (x + y, 1)$.
3. $T_3 \colon \mathcal{P}_2 \to \mathcal{P}_2, T_3(p)(t) = tp'(t)$.
4. $T_4(f) = f^2$ for $f \in C^\infty(\mathbb{R})$, with pointwise multiplication.

CH08-E2

Define $T \colon \mathbb{R}^3 \to \mathbb{R}^2$ by

$$T(x, y, z) = (x + 2y - z,\ x - y + 2z).$$

1. Find its standard matrix.
2. Find a kernel basis and nullity.
3. Find an image basis and rank.
4. Verify rank-nullity.
5. Decide injectivity and surjectivity.

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CH08-E3

Let $T\colon \mathbb{R}^2 \to \mathbb{R}^3$ have standard matrix

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Input basis:

$$B_{\text{in}} = \{(1,1)^T, (1,-1)^T\}.$$

Output basis:

$$B_{\text{out}} = \{e_1, e_2, e_3\}.$$

1. Find its matrix in the specified input basis and standard output basis.
2. If the output basis changes to

$$(1,0,0)^T, (1,1,0)^T, (1,1,1)^T,$$

find the new matrix. 3. Put the ordered input basis and the output basis in part 2 into column matrices, then verify the coordinate conversion formula.

CH08-E4

Let

$$A = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}.$$

1. Find eigenvalues and eigenvectors.
2. Construct B making $B^{-1}AB$ diagonal.
3. Compute the diagonal matrix and check trace and determinant.
4. Describe the effect of multiplying each eigenvector by a nonzero scalar.

CH08-E5

Consider derivative operator $D\colon \mathcal{P}_2 \to \mathcal{P}_1$ with monomial bases.

1. Give its coordinate matrix.
2. Find kernel and image and verify rank-nullity.
3. Change the domain basis to $1, t-1, (t-1)^2$ and find the new matrix.
4. Explain why these rectangular matrices are not similar.

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CH08-E6

Give

$$N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

1. Find the eigenvalue and eigenvectors of N.
2. Find a generalized eigenvector w whose image Nw is an ordinary nonzero eigenvector.
3. Give a Jordan form and basis matrix satisfying $N = BJB^{-1}$.
4. Explain why N is not diagonalizable despite having only one distinct eigenvalue.

CH08-E7

Fix the cyclic permutation direction as

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \qquad P(x_0, x_1, x_2, x_3)^T = (x_1, x_2, x_3, x_0)^T.$$

and consider the Fourier basis vectors

$$v_\lambda = (1, \lambda, \lambda^2, \lambda^3)^T, \qquad \lambda \in \{1, i, -1, -i\}.$$

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1. Prove $Pv_\lambda = \lambda v_\lambda$.
2. Prove independence of these four vectors.
3. For $C = c_0 I + c_1 P + c_2 P^2 + c_3 P^3$, give the diagonal entries in this Fourier basis.
4. Explain why a real P can have complex eigenvalues.

CH08-E8

Let

$$B_\epsilon = \begin{bmatrix} 1 & 1 \\ 0 & \epsilon \end{bmatrix}.$$

1. Prove the columns of B_ϵ form a basis for every $\epsilon > 0$.
2. Find $\kappa_2(B_\epsilon)$ and its limit as $\epsilon \to 0^+$.
3. Describe the angle between the basis vectors.
4. Bound reconstruction error when coordinate error is 10^{-8}. Distinguish a norm bound from a bound on each coordinate, and interpret the conditioning.

CH08-E9

Let $T(x,y)=(x+y,x-y,2x)$, $S(x,y,z)=(x+z,y-2z)$.

1. Find their standard matrices.
2. Find the standard matrix of $S\circ T$ and explain the multiplication order.
3. For $(x,y)=(1,2)$, compare direct substitution with the composition matrix.

CH08-E10

Let $D:\mathcal{P}_2\to\mathcal{P}_2$, $D(p)=p'$, with bases $\beta=(1,x,x^2)$, $\gamma=(1,1+x,1+x^2)$. Find $[D]_\beta$, the matrix P converting γ-coordinates to β-coordinates, and $[D]_\gamma=P^{-1}[D]_\beta P$. Check using direct differentiation of $[p]_\gamma=(1,2,1)^T$.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 8.

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